LECTURE 09 · REVIEW NOTES

What Is a Comparable Dollar Worth?

Valuation by multiples, followed by a short introduction to price trends and moving averages.

Luis Ceballos, PhD · FINA 402 · Fall 2026

1. Price is not the same as cheapness

A $30 stock with $1 of forward EPS trades at 30 times expected earnings. A $90 stock with $6 of forward EPS trades at 15 times expected earnings. The second costs less per dollar of expected earnings. It is not necessarily the better investment: growth, risk, capital requirements and the reliability of earnings may differ. All numerical examples in this lecture are simulated teaching cases.

P/E = Price / EPS

P/B = Price / Book value per common share

Relative valuation estimates what the market would pay if a target were priced like comparable assets. DCF estimates value under explicit cash-flow and discount-rate assumptions. Both require judgment. An entire peer group can trade at prices unsupported by fundamentals.

A multiple contains assumptions

$$P_0 = D_1 / (k - g)$$

Forward P/E = Payout ratio / (k − g)

The second equation follows from Gordon by dividing price by next-year EPS and assuming next-year dividends equal the payout ratio times next-year earnings. The model requires sustainable stable growth and k greater than g. Forward and trailing P/E have different earnings denominators. A higher multiple can reflect stronger growth, lower required return or a higher sustainable payout.

Multiple	Match and limitation
P/E	Common equity with common earnings. Negative earnings make ordinary P/E rankings unhelpful.
P/B	Common equity with common book equity. Interpret with expected ROE and asset quality.
EV/EBITDA	Enterprise value with operating earnings before D&A.; EBITDA omits investment needs and is not free cash flow.

LECTURE 09 · REVIEW NOTES · 2

2. Selecting peers and estimating a price

Use comparable businesses with similar growth, risk, profitability and reinvestment needs. An industry label alone is insufficient. Align valuation dates, forecast periods, diluted share counts and accounting adjustments. Treat leases consistently in both enterprise value and operating earnings.

Mean versus median

For peer P/Es of 14, 16 and 18, both the mean and median are 16. Including an outlier at 60 raises the mean to 27, while the median becomes (16 + 18)/2 = 17. With target forward EPS of $4, these imply $108 and $68 respectively. The median limits outlier influence but cannot repair a poorly chosen peer group.

Worked example: the equity multiple route

Implied price = Peer median P/E × Target forward EPS

Using the original median of 16 and EPS of $4, the implied price is **$64**. If the target quote is $50, implied upside is 64/50 − 1 = **28%**. This comparison is conditional on the multiple; it does not guarantee a future return.

Worked example: the enterprise multiple route

Assume comparable forward EV/EBITDA multiples of 7, 8 and 10. Target forward EBITDA is $100m, debt is $200m, excess cash is $40m and diluted common shares are 20m. No preferred shares, noncontrolling interests or other claims/assets are included.

Step	Calculation	Result
Median multiple	Middle of 7, 8 and 10	8×
Enterprise value	8 × 100	$800m
Common equity	800 − 200 + 40	$640m
Price per share	640 / 20	$32

Do not multiply EBITDA by P/E, and do not divide enterprise value directly by shares. A difference between the $64 P/E estimate and the $32 EV/EBITDA estimate is a reason to investigate peers, leverage, depreciation, margins and investment needs. It is not a reason to average the estimates automatically.

LECTURE 09 · REVIEW NOTES · 3

3. A short technical-analysis laboratory

Valuation asks what a price is supported by. Technical analysis studies patterns in past prices and trading activity. In this lecture, the five-day simple moving average is a transparent descriptive tool. It smooths recent prices and reacts with a lag; it does not establish intrinsic value or guarantee a profitable trade.

$$SMA_5 = (P_1 + P_2 + P_3 + P_4 + P_5) / 5$$

For five closes of \$40, \$41, \$42, \$41 and \$46, the average is **\$42**. The last close is above that average. This describes information available after day 5 closes. A strategy using that closing signal cannot assume an earlier execution price.

The next execution and later outcome

Suppose a hypothetical trade enters at the next session's open of \$43 and exits later at \$44. Transaction cost is 0.10% of traded value on each side. Assume one share, no dividends and split-consistent prices. The exit price is not known at the time of the signal.

$$\text{Net return} = [\text{Exit}(1 - c) / (\text{Entry}(1 + c)) - 1] \times 100\%$$

Quantity	Calculation	Result
Gross return	(44/43 − 1) × 100	2.3256%
Entry cash cost	43 × 1.001	\$43.043
Net exit proceeds	44 × 0.999	\$43.956
Net return	(43.956/43.043 − 1) × 100	2.1211%

The 0.10% cost is 0.001 in decimal form, charged separately on entry and exit. A negative answer is a loss. A small gross profit can disappear after costs; subtracting 0.20 percentage points is only an approximation to the exact formula.

How to avoid a misleading demonstration

Reveal one day at a time and make decisions before showing the next observation. Define rules before testing, reserve unseen data, and include execution costs and slippage. Avoid selecting only survivors or successful windows. One winning trade is not evidence of a reliable predictive rule.

LECTURE 09 · REVIEW NOTES · 4

4. Activities, mistakes and self-check

Activity	Required outputs	Points
1 · Relative valuation	P/E implied stock price and EV/EBITDA implied stock price	3
2 · Technical-analysis arithmetic	Five-day SMA and realized net trade return	2

Each student has distinct, repeatable simulated inputs. Activity 1 uses the median of each three-firm peer set and consistent forward fundamentals. Complete Activity 1 before Activity 2. Prices and SMA allow ±1%; net return allows ±0.05 percentage points. Both outputs must pass to complete the activity.

Only correct saved outputs appear in the instructor's section dashboard. Attempts and attendance are tracked separately. Instructor demonstrations do not change student scores. Review mode disables submissions.

Frequent errors

Error	Correction
Equating a low share price with cheapness	Scale price by a relevant fundamental and inspect comparability.
Mixing trailing and forward inputs	Use the same earnings period for target and peers.
Treating EBITDA as cash available	Remember taxes, CapEx and working-capital needs.
Dividing EV by shares	Subtract debt and add excess cash first in the simplified case.
Seeing the future in the chart	Form the signal only with data available at that time.
Entering return as a decimal	The answer box requests percent: enter 2.12, not 0.0212.

Self-check

1. Peer P/Es are 12, 15 and 30. Target EPS is $3. What price follows from the median?
2. EV is $600m, debt $150m, excess cash $30m and shares 20m. What is the price?
3. If the stock is above its five-day average, does that prove it is undervalued?

Answers: (1) 15 × 3 = **$45**. (2) (600 − 150 + 30)/20 = **$24**. (3) No. The average describes historical price behavior and does not determine intrinsic value.

Conceptual sources

Aswath Damodaran, NYU Stern: The Anatomy of a Multiple.
Fidelity: What is technical analysis?